EchoMind PRD

The Organizational Memory Operating System

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Prepared For: Japan-India Innovation Hackathon

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1. Executive Summary

Organizations globally are facing an invisible crisis: the irreversible loss of tacit expertise. When experienced engineers, operators, technicians, and domain experts retire or churn, they take decades of intuition and decision-making context with them. While modern enterprises store vast amounts of documents, manuals, and standard operating procedures (SOPs), they rarely capture the *reasoning* behind critical decisions.

EchoMind is an AI-powered Organizational Memory Operating System designed to capture, structure, preserve, and reuse human expertise. By transforming raw decisions into searchable, contextual organizational intelligence, EchoMind shifts the enterprise paradigm. Instead of asking, *"What documents do we have?"* organizations can ask, *"Have we solved a similar problem before, and why did we choose that specific solution?"*

2. Problem Statement

Currently, organizations rely on static repositories (Reports, PDFs, Emails, SOPs, Maintenance logs, Meeting notes). However, these records are fundamentally flawed because they lack:

- **Decision Rationale:** *Why* a specific path was chosen.
- **Expert Intuition:** The subtle, unwritten indicators that guide veterans.
- **Alternative Options:** What else was considered and why it was rejected.
- **Historical Context:** The environmental or business constraints at the time.

The Business Impact:

- Knowledge disappears entirely when employees exit.
- New employees inadvertently repeat historical mistakes.
- Organizations waste resources repeatedly solving the same problems from scratch.
- Critical expertise becomes dangerously siloed in a few key individuals.

Real-World Examples

- **Manufacturing:** A senior engineer fixes a machine based on subtle vibration sounds; no record explains *how* the conclusion was reached.
- **Aerospace:** An experienced technician detects degradation through minor visual indicators; the knowledge remains entirely personal.
- **Railways & Logistics:** Field operators choose repair or routing strategies based on years of localized experience; future teams cannot replicate the reasoning.

3. Product Vision & Goals

Vision Statement: To create a permanent, evolving organizational memory that preserves human expertise beyond individual careers, enabling continuous learning and consistent decision intelligence across generations of employees.

Goal Type	Description
Primary	Preserve organizational expertise by converting tacit knowledge into structured data.
Secondary	Reduce knowledge loss during employee turnover or retirement.
Secondary	Improve decision quality by surfacing historical context.
Secondary	Accelerate employee onboarding and upskilling.
Secondary	Support succession planning and improve overall operational resilience.

4. Target Personas

Persona	Demographics	Pain Points	Primary Goal
The Veteran Expert	Age: 55+ (Senior Engineer, Specialist)	Approaching retirement; decades of undocumented, specialized knowledge in their head.	To easily capture and leave behind a legacy of expertise without heavy administrative burden.
The Junior Operator	Age: 22-30 (New Hire, Junior Engineer)	Lacks practical, contextual experience; struggles to find past solutions in messy company wikis.	To quickly learn from historical cases and make safe, accurate decisions.

Persona	Demographics	Pain Points	Primary Goal
The Ops Manager	Age: 35-50 (Department Head)	Frustrated by repeated organizational mistakes and inconsistent troubleshooting.	To improve decision consistency, reduce downtime, and monitor team knowledge gaps.

5. Core Concept: The Knowledge Architecture

Traditional systems store *information*. EchoMind stores *reasoning*. Every important organizational event is mapped using the **EchoMind Decision Framework**:

Situation → Analysis → Options Considered → Decision Taken → Outcome → Lessons Learned

This structured flow ensures that data is automatically converted into a dynamic, reusable knowledge graph rather than a flat document.

6. Functional Requirements

Feature 1: Decision Capture Engine

Description: The primary input interface for users to log events, problems, and decisions.

User Story: As an expert, I want to easily input the context and reasoning behind a decision I just made, so that the organization has a record of my logic.

Acceptance Criteria:

- Form includes fields for: Problem Description, Context/Data, Options, Decision, Reasoning, and Outcome.
- Supports text, voice-to-text, and file attachments.

Feature 2: AI Reasoning Extractor

Description: An LLM-powered engine that ingests unstructured data (meeting notes, logs) and structures it into the EchoMind Decision Framework.

User Story: As an Operations Manager, I want to upload a post-mortem report so the AI can automatically extract the key decisions and lessons learned.

Acceptance Criteria:

- System accepts .pdf, .docx, and .txt files.
- AI accurately identifies and tags the Situation, Decision, and Outcome.

Feature 3: Organizational Knowledge Graph & Timeline Explorer

Description: A visual mapping tool showing how problems, decisions, and outcomes relate to one another over time.

Acceptance Criteria: Interactive UI displaying nodes (Problems) and edges (Decisions/Outcomes) with a chronological timeline slider.

Feature 4: Similar Case Finder & Recommendation Engine

Description: The core retrieval system. Users describe a current problem, and the system finds historical matches and recommends actions.

Acceptance Criteria: Search bar accepts natural language queries. Results display similar historical situations, previous solutions, and a calculated success rate. Generates a recommended action plan.

Feature 5: Expert Knowledge Vault

Description: A dedicated, searchable repository for deep-dive expert interviews, audio snippets, and specialized case studies.

Feature 6: Organizational Learning Dashboard

Description: An analytics dashboard for leadership to track system usage and knowledge retention.

7. User Journey (Demo Flow)

1. **Trigger:** User encounters a problem (e.g., abnormal machine behavior).
2. **Query:** User asks EchoMind, "Have we seen this before?" via natural language.
3. **Retrieval:** EchoMind searches the Organizational Knowledge Graph.
4. **Recommendation:** EchoMind presents historical solutions, risk analysis, and success rates.
5. **Action & Capture:** User applies the fix, then updates EchoMind with the specific context, the decision taken, and the final outcome.
6. **Synthesis:** The AI Reasoning Extractor updates the Knowledge Graph, making the organization smarter for the next user.

8. Non-Functional Requirements

Security & Privacy

- **End-to-End Encryption:** All data at rest and in transit must be encrypted (AES-256).
- **Role-Based Access Control (RBAC):** Restrict sensitive decisions to authorized clearance levels.

- **Anonymization:** Optional toggle to remove PII from historical cases.
- **Audit Logging:** Track who accesses and edits decision records.

Scalability & Availability

- **Volume:** Architecture must support the ingestion and querying of 100,000+ complex decision nodes.
- **Uptime:** Target 99.9% availability for critical operational environments.

9. Success Metrics (KPIs)

Metric	Definition	Target
Knowledge Retention Rate	% of critical operations documented vs. expert churn.	80%
Decision Reuse Rate	% of new problems solved using historical EchoMind data.	60%
Onboarding Time Reduction	Decrease in time-to-productivity for new hires.	40%
Repeated Mistake Reduction	Drop in recurring errors/downtime incidents.	30%

10. Competitive Advantage

Feature/ Focus	Traditional KM (Wikis/ SharePoint)	Traditional AI (ChatGPT)	EchoMind
Core Asset	Stores static documents	Answers general questions	**Stores organizational reasoning**
Primary Value	Archival	Immediate general knowledge	**Contextual organizational memory**
Output Type	Search results (files)	Synthesized text	**Actionable, historically proven decisions**

11. Future Roadmap

- **Phase 1: Decision Capture Platform (MVP)** - Core input forms, manual data entry, basic search.
- **Phase 2: Knowledge Graph Integration** - Implementation of the visual node-based graph and AI Reasoning Extractor.
- **Phase 3: AI Recommendation Engine** - Predictive analytics, automated risk scoring, and natural language matching.

- **Phase 4: Cross-Organization Learning Network** - Federated learning between non-competing organizations.

12. Hackathon Demo Scenario

The Setup: A junior engineer is on the factory floor and encounters an abnormal, high-frequency machine vibration on Conveyor Belt B. The senior engineer who usually handles this retired last month.

The Action: The user opens EchoMind and types: *"Have we seen abnormal high-frequency vibration on Conveyor B before?"*

The EchoMind Response:

- **Status:** "Searching Organizational Memory..."
- **Results Found:** 14 similar cases found in the last 5 years.
- **Historical Breakdown:**
  - *11 cases* resolved through bearing replacement (Success Rate: High).
  - *2 cases* resolved through lubrication correction.
  - *1 case* resolved through shaft alignment.
- **AI Recommendation:** "Based on historical reasoning from [Retired Expert Name], check the aft bearing temperatures first. If above 180°C, replace immediately to prevent total motor failure."

The Outcome: The junior engineer safely resolves the issue in 20 minutes instead of 4 hours. The organization benefits from decades of accumulated expertise, even though the original expert is no longer in the building.

"People leave. Knowledge shouldn't."